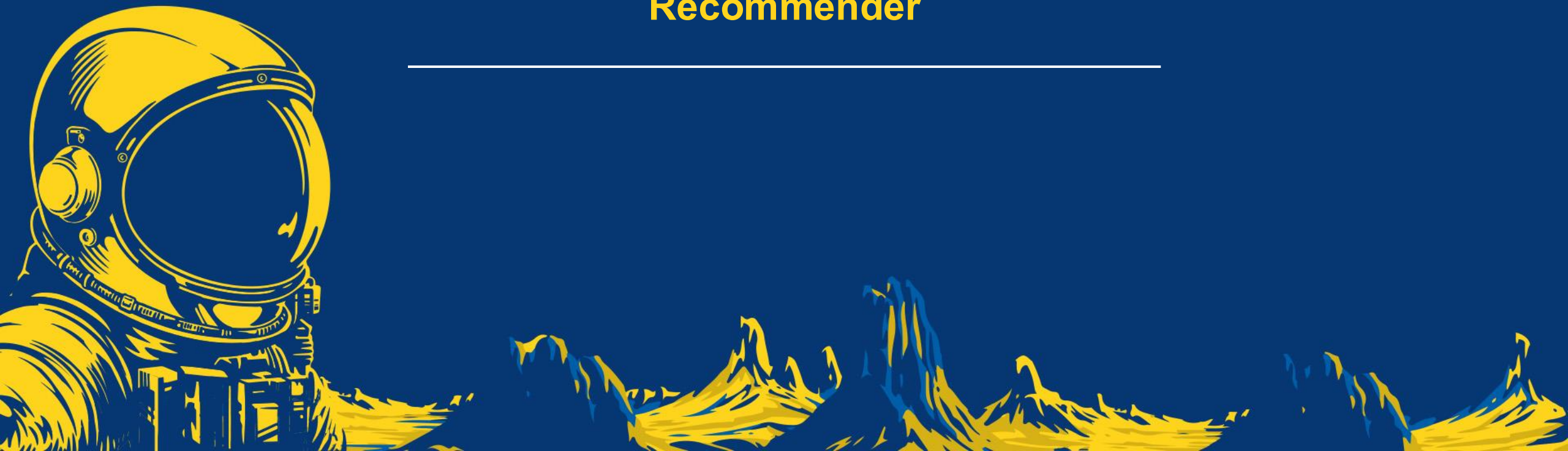




Building an AI-based Hotel Recommender



There are many use cases at CHECK24 where AI brings great added value

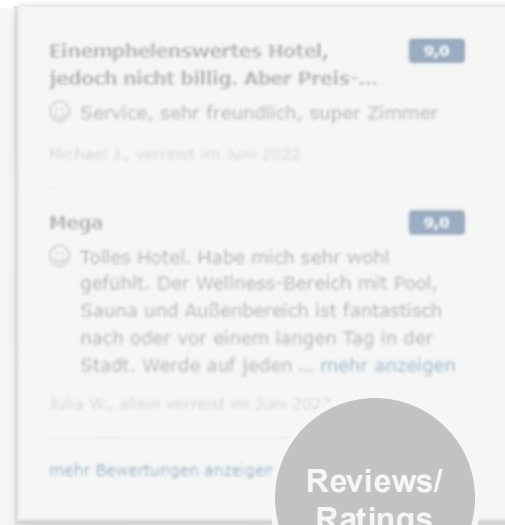
Chatbots



E-Mails



Reviews/
Ratings



Documents



Trans-
actions



Ranking &
Recommendations



Images



Your mission: Build an AI-powered hotel recommendation system

1

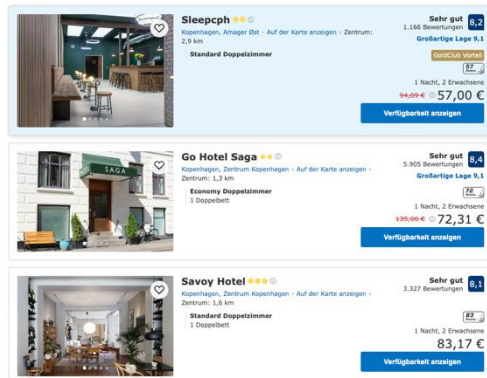
"I'm looking for a family-friendly hotel that has a nice indoor pool and a relaxing sauna where we can unwind after a long day"

Your AI System



- [Max. 10 Hotels]
- "No hotels found"
- "Invalid request"

2 100 Pre-ranked hotels



Your system should be able to handle three distinct types of User Queries

① Prompts with fully or partially matching hotels

"I'm looking for a hotel where I can enjoy a lovely dinner, as well as a relaxing experience at a spa and wellness center. It would also be great if the hotel has an elevator to make getting around easier. Could you help me find a place that fits these needs?"



**[List Max. 10 Hotels
sorted in descending
order]**

② Prompts with no matching hotels

"5 stars hotels with 20 euro budget per night"



[empty list]

③ Irrelevant Prompt

"Can you recommend a good strategy for winning Monopoly?"



NaN

Step by step process

1. One member creates a team on GitHub

- Open the assignment link → Log in with GitHub → Create team → Accept assignment.

2. Others join the team

- Open the same link → Log in → Join team → Accept assignment.

3. Coding

- Create your recommender system with function *find_matching_hotel()* at its core

4. Interim Evaluation (optional):

- Push your code till **Saturday at 4:00 PM**

5. Final submission by pushing the code (until Sunday, April 27th at 10:00 AM)

- Final solution = Code pushed to your team's private repo

6. Evaluation

- We will run 50 test cases (different destinations and user prompts) through your function

Additional Resources: You get access to OpenAI Models and some examples of the hotel data and user queries



Available models:

- o3-mini
- gpt-4o
- gpt-4o mini



Hotel lists from 3 Cities with all the features and amenities

- Copenhagen
- Mallorca
- New York



Some user prompt examples



Important: API key credentials, along with usage instructions, will be provided in the **repo**.



Hotel lists from 3 Cities: 100 Pre-ranked Hotels

List of hotels with detailed attributes such as amenities, price, rating etc. for 3 different destinations

Sleepcph ★★★ 📍

Kopenhagen, Amager Øst - Auf der Karte anzeigen - Zentrum: 2,9 km

Standard Doppelzimmer

Sehr gut 8,2
1.166 Bewertungen

Großartige Lage 9,1

GoldClub Vorteil

1 Nacht, 2 Erwachsene
~~94,99 €~~ 📉 57,00 €

Verfügbarkeit anzeigen

Go Hotel Saga ★★★ 📍

Kopenhagen, Zentrum Kopenhagen - Auf der Karte anzeigen - Zentrum: 1,3 km

Economy Doppelzimmer
1 Doppelbett

Sehr gut 8,4
5.905 Bewertungen

Großartige Lage 9,1

1 Nacht, 2 Erwachsene
~~935,06 €~~ 📉 72,31 €

Verfügbarkeit anzeigen

Savoy Hotel ★★★★ 📍

Kopenhagen, Zentrum Kopenhagen - Auf der Karte anzeigen - Zentrum: 1,6 km

Standard Doppelzimmer
1 Doppelbett

Sehr gut 8,1
3.327 Bewertungen

1 Nacht, 2 Erwachsene
83,17 €

Verfügbarkeit anzeigen

searchid	clustername	clusterchain	clusterbrand	clusterabbr	bookingshar	pricepernight	rating	ratingscount	starcateg	mealtype	cancelable	ncetorefer	stancetocic	ncetounder	stancetobeg
394618531	Luna Blanca Resort & Spa - All Inclusive				2,08	102,44	9	1438	5	all_inclusiv	FALSE	5,46	5,46		0,489
394618531	Side Crown Palace Ultra All Inclusive				4,02	144,67	8,3	3432	5	all_inclusiv	FALSE	3,47	3,47		0,445
394618531	Hane Sun Elite Hotel				0,28	106	9,1	412	5	all_inclusiv	FALSE	7,65	7,65		0,871
394618531	J'adore Deluxe Hotel & Spa				1,38	123,33	8,8	2569	5	all_inclusiv	FALSE	6,55	6,55		
394618531	Adalya Art Side - All Inclusive				0,74	173,56	9,6	1466	5	all_inclusiv	FALSE	6,83	6,83		0,521
394618531	Aydinbey King's Palace & Spa - All Inclusive				1,83	206,76	9,2	2308	5	all_inclusiv	FALSE	6,02	6,02		0,176
394618531	Port River Hotel - All Inclusive				0,28	142	8,4	1048	5	all_inclusiv	FALSE	6,45	6,45		
394618531	Oz Hotels Side Premium				1,31	152	8,9	2323	5	all_inclusiv	FALSE	6,58	6,58		0,392

Submission Guidelines



Please make sure to **carefully read the README** and follow **all specified requirements** (e.g., Python version, code structure) to ensure your solution can be properly evaluated.



Submission Instructions: Push your code to the **GitHub repository** that was automatically created when you accepted the assignment.



Submission Deadline: Sunday, April 27 before 10:00 AM



Additional Deliverables



- **Creative name** for your AI system
- **Short presentation** including a short hands-on demo



How your solution will be judged



Accuracy (70%): Correctness of predictions (**DCG based on test recommendation**)



Runtime (10%): Average response time of prediction. The score decreases linearly with slower response times.



Creativity (10%): Originality and elegance of the solution handling various situations as well as edge cases.



Presentation (10%): Clarity and structure of the presentation.



Each dimension is scored from **0 to 1**.

The **final score** is the **weighted average** of these **individual scores**.

Example to calculate accuracy

Test Set 1		
Test Answer	Your Submission	Accuracy Score (NDCG)
[A, B, C]	[A, B, C]	1
[A, B, C]	[C, B, A]	0.78
[A, B, C]	[D, E, G]	0

Interim Evaluation: Leaderboard Update – Saturday at 4:00 PM

Hotel Matcher Leaderboard

Top Performers



codespy

0.427



super_donut

0.234



lobby_llamas

0.131

Detailed Scores

Team	 Final Score	 Hotel Match	 No Match	 Invalid Cases	 Runtime	overall_score
codespy	0.427	0.279	0.417	1.000	0.748	0.426683
super_donut	0.234	0.035	0.250	1.000	0.647	0.233887
lobby_llamas	0.131	0.197	0.000	0.000	0.000	0.131162

Your Prizes

4 x



4 x



4 x



2

1

3

Let's Get Started!

- You need a **GitHub account** to access the task and submit your solution.
- Please go to the **[assignment link](#)** to **create your team** and accept the assignment.
- A new **private repo is automatically** created for the team in which you can work together.
- After the team is created, it will appear in the **list of the available teams** and all team members can join it following the **same [assignment link](#)**.



Important: The **assignment link** will **expire on Saturday at 10 AM!**
We will also post this description in the Discord channel!

Happy coding!

